Functional Mappings

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In these notes, we will discuss the principles of injective, surjective, and bijective functions and their enthralling range of applications.

Injections and Surjections

Injective

We call a function $f : X \to Y$ *injective* if f maps distinct elements from X to distinct elements in Y. In other words, if $x_1 \neq x_2$, then $f(x_1) \neq f(x_2)$.

We call X the domain of f and Y its codomain. Note that f's range is a subset of its codomain, since range is simply the values in Y that $f(x)$ for $x \in X$ can achieve. This suggests that not every single value in Y can be attained by mapping a value from X.

One way to think about this is relationships: no, not math relationships, actual real-life relationships. Consider two friend groups A and B in which all members in A are dating someone from B. Thus, we can write a mapping $g : A \to B$ that maps one person in A to someone in B they are dating. The mapping g should optimally be an an actual function because things get really bad when one person in A is dating two people in B. It is hopefully injective as well, which would mean that one person in B is only dating up to one person in A.

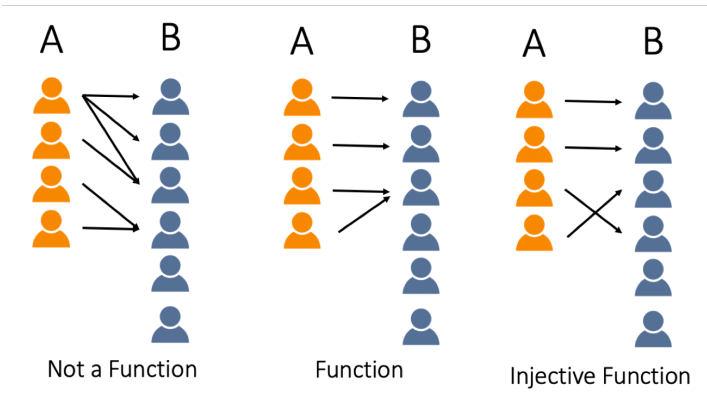

Figure 1: Injectivity Visualized for $f : A \to B$

Note that this doesn't mean that everyone in B is dating someone in A. If we want to ensure this property, we need a new condition for $g : A \to B$.

Surjective

A function $f : X \to Y$ is called *surjective* if every single value in Y can be attained by mapping a value x from the domain X, which produces $f(x) = y$. Note that multiple values of $x \in X$ can map to a same value $y \in Y$ as long as the function is not injective as well.

We can once again consider the relationship metaphor. If the dating function $f : A \to B$ is surjective, this means that everyone in B has at least one partner in A. However, f is not injective, then someone in B could possibly have more than one partner in A, which is problematic. A function that is both surjective and injective is called *bijective*.

Bijection

A function $f : X \to Y$ is called a *bijection* if it is both injective and surjective. A bijection is called one-to-one correspondence.

Note that if X and Y are both finite sets in bijection $f : A \to B$, then the number of elements in X, denoted $|X|$, is equal to the number of elements in Y, denoted $|Y|$. Additionally, each element $x \in X$ maps to a singular and unique element y in Y. In the relationship example, this means that within the equally sized friend groups A and B, each person in A is dating a unique person in B.

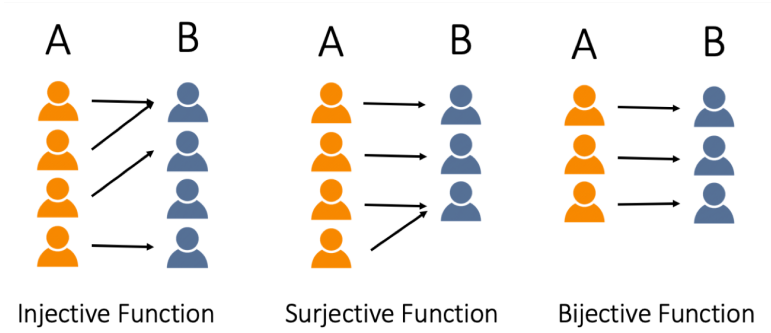

Figure 2: Surjectivity and Bijectivity Visualized for $f : A \to B$

It turns out that the injective, surjective, and bijective properties of functions have applications that extend beyond relationships. We will explore a few problems involving the use of these properties in relation to combinatorics.

Example 1

How many surjective functions $f : A \to B$ map from the set $A = \{1, 2, 3, 4\}$ to the set $B = \{1, 2, 3\}$?

Solution: The definition of a surjective function is that each element in B has at least one element in A. Thus, we can partition the set A into $|B| = 3$ subsets, where in each subset, every element maps to a singular element in B. A such partition would be $\{1, 4\}$,$\{2\}$, and $\{3\}$. The number of possible partitions is the same as the number of ways to select a distinct two-element subset (as the other subsets must each have one element), which can occur in $\binom{4}{2} = 6$. Then, each subset is assigned to a distinct element in B, which can occur in $3! = 3 * 2 * 1 = 6$ ways, as there are 3 choices in B for the first subset, 2 choices for the second subset, and 1 remaining choice for the final subset. Thus, the total number of surjective mappings is $6 * 6 = 36$.

In the left surjection below, the elements in $\{1, 4\}$ both map to $1 \in B$, the element in $\{2\}$ maps to $2 \in B$ and the element in $\{3\}$ maps to $3 \in B$.

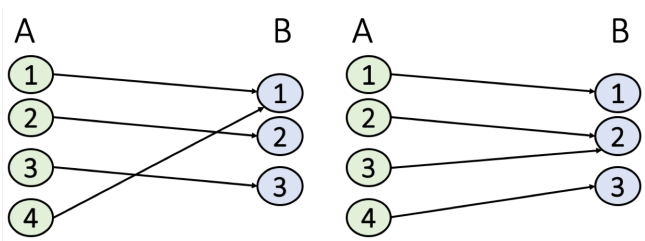

Figure 3: Two sample surjections $f : A \to B$.

Example 2

How many injective functions $f : A \to B$ map from the set $A = \{1, 2, 3, 4\}$ to the set $B = \{1, 2, 3, 4, 5\}$?

Solution: The definition of an injective function is that each element in A maps to a unique element in B. Since there are 4 elements in A, there are 4 elements in B that an element in A can map to. There are $\binom{5}{4} = 5$ number of ways of selecting these elements. Next, we consider the number of ways to assign each of the elements in A to one of the four selected elements in B. This can transpire in $4! = 4 * 3 * 2 * 1$ ways, as there are four options for the first element in A, three options for the next element, and so forth, with one remaining option for the last element. Thus, the total number of injective functions is $4! * 5 = 24 * 5 = \boxed{120}$.

Generalized Function Counting and Cryptology

The next conceptual step is to generalize Problems 4 and Problem 5. Let A and B be finite sets and consider a function $f : A \to B$ between the two sets. Also consider the number of elements in each set, which can be denoted as $|A|$ and $|B|$ for sets A and B, respectively.

- If $|A| \geq |B|$, then f can be a surjective mapping. The number of possible surjective mappings is $|B|! * S(|A|, |B|)$, where $S(|A|, |B|)$ is the number of ways to partition an $|A|$-element set into $|B|$ non-empty subsets. A more concrete equation purely involving $|A|$ and $|B|$ is quite exotic and for now beyond the scope of this course, but the logic follows a similar thought process as Example 1 except with variable and not fixed set sizes.

- if $|A| \leq |B|$, then f can be an injective mapping. The number of possible injective mappings is $\binom{|A|}{|B|} * (|B|!) = \binom{|A|}{(|A|-|B|)!}$. The proof follows a similar thought process as Example 2 except with variable and not fixed set sizes.

- If $f : A \to B$ is both surjective and injective—and by extension, a bijection—then $|A| = |B|$ must hold true. The number of possible bijections is $|A|! = |B|!$.

The proof is relatively simple for the bijection case. Let $n = |A| = |B|$. We note that each element in A corresponds with one unique element in B and vice versa. For the first element in A, there are $n = |B|$ possible elements in B to map to. For the next element, there are $n - 1$ possible selections, and so forth, with the last element having just one choice. The total number of bijections is thus $n * (n - 1) * (n - 2) * ... * 1 = n!$.

Example 5

In English, a simple substitution cipher is a mapping $f : A \to A$ in which each letter in the alphabet $A = \{a, b, c, ..., z\}$ is mapped to a unique letter in $A = \{a, b, c, ..., z\}$. These are commonly used as methods of encryption and hiding messages. For instance, if $a \to s$, $m \to a$, and $c \to d$ are part of a cipher, then the message *amc* can be encrypted into *sad*. How many distinct simple substitution ciphers are there in the English alphabet?

Solution: This topic arises from cryptology, a branch of math associated with creating and breaking codes. We can represent any simple substitution cipher as a permutation of the 26 letters in the alphabet. For instance, a sample cipher is $dbktlxefzarijcwymusopvqhng$, where a maps to d, b maps to b, c maps to k, d maps to t, and so forth, with z mapping to g. We also note that any permutation of the 26 letters describes a unique cipher. Therefore, there is a *bijection* between the set of simple substitution ciphers and the set of 26-letter permutations.

Let's count the number of 26-letter permutations. We have 26 choices for the first letter, 25 choices for the second letter, 24 choices for the third letter, and so forth, culminating in a grand total of $26 * 25 * 24 * ... * 1 = \boxed{26!}$ possible 26-letter permutations and thus simple substitution ciphers. A visualization of a substitution cipher is shown below for visual reference.

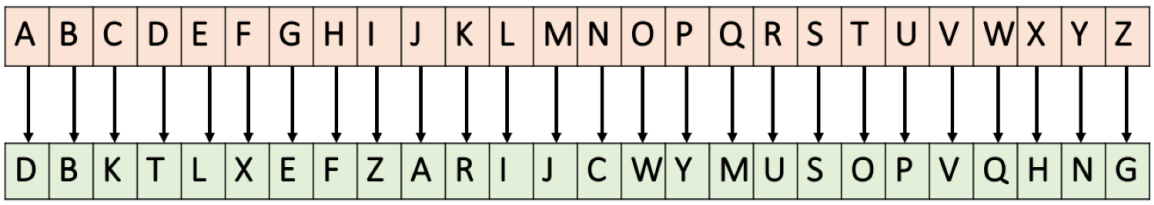

Figure 4: A simple substitution cipher that maps from each letter in red to another letter in green. Using this cipher, the message *abcd* would become *dbkt*.

Exercises

1. Let A and B be finite sets and let $|A|$ and $|B|$ be the number of elements in A and B, respectively. Prove that if $|A| < |B|$, there exists no surjective mapping $f : A \to B$.

2. Let A and B be finite sets and let $|A|$ and $|B|$ be the number of elements in A and B, respectively. Prove that if $|A| > |B|$, there exists no *injective* mapping $f : A \to B$.

3. Determine which of the following examples are injective functions.

   - $f(x) = x^2 + 2$ for x in the domain $[-\infty, \infty]$.
   - $f(x) = x^2 + 2$ for x in the domain $[0, \infty]$.
   - $f(x) = \sqrt{x}$ for x in the domain $[0, \infty]$.
   - $f(x) = \sin(x)$ for x in the domain $[-\frac{\pi}{2}, \frac{\pi}{2}]$.

4. For each of the bijections $f : A \to B$ below, determine a reasonable mapping strategy from an element in one set to an element in the other set. Consider how one may build logical pairs of corresponding elements between sets.

   - $f : A \to B$. A is the set of all paths from $(0, 0)$ to $(4, 4)$ moving only one unit up (U) or right (R) at a time. B is the set of all letter arrangements of $RRRRUUUU$.
   - $f : A \to B$. A is the set of all intersection points of the diagonals within an n-sided polygon where no three diagonals are concurrent (meet at the same point). B is the set of all combinations of 4 vertices of the n-sided polygon.
   - $f : A \to B$. A is the number of sequences of 4 dice rolls where each of the final three rolls is at least as large as the roll preceding it. B is the number of ways to move from $(0, 0)$ to $(4, 5)$ moving only one unit up or right at a time. (related to AIME I 2001)

5. Let S be the set of simple substitution ciphers. If a random cipher from S is selected, what is the probability that each letter maps to a different letter? In other words, $f(x_i) \neq x_i$ for all letters of the alphabet x_i and cipher f.

6. A special type of simple substitution cipher is a Caesar cipher, where each letter is "shifted" k numbers to the right, where k ranges between 0 and 25. For instance, if $k = 1$, then a would be shifted to b—in other words, $f(a) = b$. Similarly, $f(b) = c$, $f(c) = d$, and so forth. If $k = 3$, then $f(a) = d$, $f(b) = e$, and so forth. Note that f^{-1} describes the decryption function, which reverses the encryption.

The following text is written in cipher text, meaning that it has already been processed through a Caesar shift of 4 units right. Reverse the shift to determine the actual underlying message.

PMR GERYIP QMVERHE

7. Does there exist a bijection f between the set of positive integers and the set of positive *even* integers? If so, we say that the cardinality, which describes the number of elements, of the two sets is equal.

8. Let $(a_1, a_2, a_3, \ldots, a_{12})$ be a permutation of $(1, 2, 3, \ldots, 12)$ for which

$$a_1 > a_2 > a_3 > a_4 > a_5 > a_6 \text{ and } a_6 < a_7 < a_8 < a_9 < a_{10} < a_{11} < a_{12}.$$

   An example of such a permutation is $(6, 5, 4, 3, 2, 1, 7, 8, 9, 10, 11, 12)$. Find the number of such permutations. (AIME II 2006)

9. A frog is positioned at the origin of the coordinate plane. From the point (x, y), the frog can jump to any of the points $(x + 1, y)$, $(x + 2, y)$, $(x, y + 1)$, or $(x, y + 2)$. Find the number of distinct sequences of jumps in which the frog begins at $(0, 0)$ and ends at $(4, 4)$. (AIME II 2018)

10. Let $A = \{1, 2, 3, 4, 5, 6, 7\}$, and let N be the number of functions f from set A to set A such that $f(f(x))$ is a constant function. Find the remainder when N is divided by 1000. (AIME II 2013)

11. Let $A = \{1, 2, 3, 4\}$, and f and g be randomly chosen (not necessarily distinct) functions from A to A. The probability that the range of f and the range of g are disjoint is $\frac{m}{n}$, where m and n are relatively prime positive integers. Find m. (AIME I 2014)

12. Call a set of integers spacy if it contains no more than one out of any three consecutive integers. How many subsets of $\{1, 2, 3, \ldots, 12\}$, including the empty set, are spacy? (AMC 12A 2007)